# Phase 2: Morphological Visualization Engine — Implementation Plan v4.0

**Document ID:** IMPL-PHASE2-REV4.0

**Supersedes:** v3.0 (2026-04-27) + v3.1 Addendum (2026-04-27)

**Author:** Kytheion (IC-004-R-D-KYN)

**Date:** 2026-04-27

**Protocols:** CP-002-O-D-JNP v2.0 + CP-010-O-D-ENK v1.3 + Tesla Principle + Fingerprint-Mugshot Pattern + Layered Compositing Model

**Gate:** Verbatim approval required: "i approve of your plan" AND "you may begin."

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## Revision Summary

This revision **unifies** v3.0 and v3.1 into a single document and adds three substantive expansions based on DIMA UX directives (2026-04-27):

| Section | v3.0/v3.1 | v4.0 |
|---|---|---|
| Results Layout | Dual-render viewport + telemetry sidebar | **Component Breakdown Dashboard** — every TEM module gets its own mini-visualization + stats row |
| Detail View | Not present | **Step 2D: Focus Lens** — contextual 2.5D high-res wireframe of selected byte region |
| Graph Display | Separate tab/toggle (Step 3) | Static 3D graph thumbnails **inline** next to their respective component rows |
| Design Language | Unspecified beyond Liquid Glass | **Full color specification** extracted from 13 reference images |
| Information Density | Implicit | **Explicit Tufte/Bloomberg principles** — every pixel carries data |

Steps 0, 1, 2A, 2B, 2C, 3, 5, 6, 7 carry forward from v3.0. Step 7B carries from v3.1. All math, bindings, falsifiability gates, and governance acknowledgments unchanged.

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# Visual Reference Index

All reference images in C:\QuarantineZone\shape-scan\plan-implimentation\:

| Image | Content | Design Tokens Extracted |
|---|---|---|
| **IMG_2859** | NotebookLM dual-render concept — torus primary, stellated thumbnail | Layout proportions, dual-viewport positioning |
| **IMG_2864** | Master architecture infographic — full pipeline from binary to 3D | Morphology bindings data table, pipeline flow, color palette |
| **IMG_2865** | Binary File Analysis Thermodynamic Profile — superformula with annotations | Leader-line style, entropy heat overlay (orange-red), wireframe (cyan-blue), annotation panel glass |
| **IMG_2867** | The Shape-Scan Pipeline — 6-stage module icons | Module iconography, stage-to-stage flow aesthetic, background particle field |
| **IMG_2868** | Dual-Channel Cognitive UI — wireframe mesh primary + ambient thumbnail | 2.5D wireframe detail render style, peak/valley topology rendering |
| **IMG_2869** | The Superformula Mugshot [Experimental] — Voronoi/ Markov projection | Equatorial band + polar cap tessellation visual, orange-red heat cells on blue wireframe |
| **IMG_2870** | The Morphological Encyclopedia — Threat Atlas 3×2 grid | Small-multiples grid layout, glass card per morphology, category color coding |
| **IMG_2871** | Morphology Profile: Clean Text (.txt) — smooth torus + linear chain graph | Benign color palette (soft blue-violet, no red), graph as simple chain |
| **IMG_2872** | Morphology Profile: PE Binary (.exe) — spiked torus + clustered graph | Threat color palette (red-orange spikes on cyan wireframe), dense graph clusters |
| **IMG_2873** | Morphology Profile: Encrypted Payload (.aes) — perfect donut + point cloud | Amber/gold for encrypted, uniform smooth surface, diffuse point cloud graph |
| **IMG_2874** | The Zugzwang Visualization — Venn of entropy/topology/ compression | Multi-view contradiction-as-signal concept, overlapping detection domains |

| | | |
|---|---|---|
| **IMG_2875** | Operator Amplification — all component panels orbiting central renders | **Master layout reference** — shows ALL component panels arranged around the 3D core |

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# Design Language Specification

### Color Palette (extracted from reference images)

| Token | Hex | Usage |
|---|---|---|
| --bg-deep | #0a0e17 | Application background, deepest layer |
| --bg-panel | #111827 | Panel backgrounds, card surfaces |
| --bg-glass | rgba(17, 24, 39, 0.85) | Glassmorphism panels with backdrop-blur |
| --wire-primary | #22d3ee | Primary wireframe mesh (cyan) |
| --wire-secondary | #3b82f6 | Secondary wireframe, grid lines |
| --heat-low | #1e40af | Low-entropy regions (deep blue) |
| --heat-mid | #f59e0b | Medium-entropy regions (amber) |
| --heat-high | #ef4444 | High-entropy spikes (red) |
| --heat-critical | #ff6b35 | Critical anomaly glow (orange-red) |
| --benign | #8b5cf6 | Clean/benign file morphology (violet) |
| --encrypted | #d97706 | Encrypted payload (amber/gold) |
| --executable | #ef4444 | Binary/executable threat (red) |
| --text-primary | #e2e8f0 | Primary text, stat values |
| --text-secondary | #94a3b8 | Labels, descriptions |
| --text-mono | #67e8f9 | Monospace readouts, hashes |
| --accent-glow | #06b6d4 | Particle effects, connection lines |
| --border-glass | rgba(148, 163, 184, 0.2) | Panel borders, dividers |
| --spark-line | #22d3ee | Sparkline stroke |

| | | |
|---|---|---|
| --spark-threshold | #ef4444 | Entropy >7.5 threshold line |

**Typography**

| Role | Font | Size | Weight |
|---|---|---|---|
| Headings | 'Inter', sans-serif | 14-18px | 600 |
| Body text | 'Inter', sans-serif | 12-13px | 400 |
| Stat values | 'JetBrains Mono', monospace | 13-16px | 500 |
| Labels | 'Inter', sans-serif | 10-11px | 500, uppercase, letter-spacing: 0.05em |
| Annotations | 'JetBrains Mono', monospace | 11px | 400 |

## Shape Language

- **Panel corners:** 8px radius (glass cards), 4px radius (inline stats)
- **Borders:** 1px solid --border-glass, no heavy borders
- **Glass effect:** backdrop-filter: blur(12px), semi-transparent background
- **Glow:** box-shadow: 0 0 20px rgba(6, 182, 212, 0.15) on active/focused elements

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# Resolved Open Questions

| OQ | Resolution |
|---|---|
| **OQ-4** | **70/30 side-by-side split** confirmed by DIMA. Thumbnail to the right. |
| **OQ-5** | Structural anomaly index weights $w_1=0.4$, $w_2=0.4$, $w_3=0.2$ accepted as starting point. Empirical tuning during Step 7B. |
| **OQ-6** (NEW, RESOLVED) | Static 3D graph thumbnails placed **inline** next to their respective component rows in the breakdown panel. |
| **OQ-7** (NEW, RESOLVED) | Detail lens (Step 2D) renders as contextual 2.5D wireframe, cross-linked with all other views. |

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# Step-by-Step Execution Order

## Step 0: TFEA Extension

**Priority:** Critical — prerequisite for Step 2

Unchanged from v3.0. Two scalar fields added to TEMReport:

- compression_ratio: f64 — DEFLATE-derived structural redundancy

- structured_fraction: f64 — proportion of file bytes covered by TCGE basic blocks

**Files:** src/tfea.rs [MODIFY], src/pipeline.rs [MODIFY]

**Scope:** ~50 lines Rust, ~30 lines test

**Gate:** TFEA tests pass; Phase 0 verdict invariance confirmed.

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# Step 1: Viewport Interaction Controls

**Priority:** Critical — unblocks all visual work

**File:** ui/js/controls.js [NEW]

Mouse wheel zoom, click-drag rotate, right-click pan, trackpad gestures, double-click reset. Controls bind dynamically to whichever render holds primary status (re-targeted by render_swap.js on swap). Thumbnail explicitly excluded from all input.

**Verify:** Zoom/pan/rotate primary via all methods. Thumbnail unresponsive. Double-click resets primary only.

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# Step 2A: Toroidal Markov Surface — Fingerprint View

**Priority:** Critical — canonical mathematical reference

**File:** ui/js/markov_torus.js [NEW] (replaces markov.js)

Parametric torus mesh, 128×128 vertices (~16K). All bindings [Established]:

| TEM Field | Surface Parameter | Confidence |
|---|---|---|
| Markov P(v\|u) | Minor radius: `r(u,v) = 0.3 + 0.5·P(v | u)^1.5` |
| Per-row entropy $H_i$ | Roughness amplitude: r_amp = 1.0 + λ·$H_i$_norm | Established |
| Markov gradient | Surface displacement: `disp = κ·( | ∂P/∂u |
| AISE intent | Crease angle: θ = π·(1-I_AISE) | Established |

**Color mapping:** per-vertex probToColor(P(v|u)) using the --heat-low → --heat-mid → --heat-high gradient.

**Visual targets** (from reference images):

- Clean text (.txt) → smooth blue-violet ring, slight ASCII ridges (IMG_2871)
- Encrypted (.aes) → near-perfect amber/gold donut (IMG_2873)
- PE binary (.exe) → asymmetric red-orange spikes on cyan wireframe (IMG_2872)

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# Step 2B: Superformula Morphology Engine — Mugshot View

**Priority:** Critical — operator-facing categorical identity

**Files:** ui/js/morphology.js [NEW], ui/js/markov_projection.js [NEW]

**2B.1: Base Mesh** — 3D Gielis spherical product, 128×128 vertices.

**2B.2: Parameter Bindings** — Layer 1/2/3 from v3.0:

| TEM Field | Superformula Param | Visual Result |
|---|---|---|
| Shannon entropy (global) | $m_1$ (symmetry order) | Lobe count |
| Per-row entropy variance | $m_2$ | Latitudinal frequency |
| Compression ratio | $a_1/a_2$ | Global shape distortion |
| Structured fraction | $n_{11}/n_{21}$ | Smooth (data) vs spiky (intent) |
| Header mismatch | Asymmetry injection | Localized deformation |
| AISE intent | Crease angle | Dihedral sharpness |
| AISE flags | Localized deformation kernels | Regional distortion |
| Entry point | Emissive light source | Glowing hotspot |

**2B.3: Hybrid Markov Projection** — Equatorial band ($\phi \in [-\pi/4, \pi/4]$) parameterized by Markov coordinates. Polar caps tessellated by Voronoi cells seeded by top-8 highest-entropy rows. Cosine blending kernel width $\delta = \pi/16$. (Visual reference: IMG_2869)

**2B.4: Validation Gates** — C4 (six-class distinguishability) + C6 (seam handling). Torus fingerprint as fallback.

**Visual targets** (from reference images):
- The "Encrypted Donut" — near-sphere with smooth contours (IMG_2864 bottom-left)
- The "ASCII Ridged Ring" — organic, moderate-complexity form (IMG_2864 bottom-center)
- The "Creature" — asymmetric, fractal-like, knotted (IMG_2864 bottom-right, IMG_2870 threat atlas)

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## Step 2C: Dual-Render UI Layout

**Priority:** Critical — operator interface

**File:** ui/js/render_swap.js [NEW]

**Layout:** 70/30 side-by-side split. Primary (70%) left, thumbnail (30%) right. Default: superformula primary, torus thumbnail.

**Swap:** Click thumbnail → 300ms animated transition → positions swap, controls re-bind.

Liquid Glass smooth animation.

**Thumbnail auto-rotation:** ~6 RPM, independent of primary interaction. Only responds to swap-trigger click.

**Visual reference:** IMG_2859, IMG_2868.

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## Step 2D (NEW): Focus Lens — Contextual Detail View

**Priority:** High — operator inspection tool

**File:** ui/js/detail_lens.js [NEW]

The Focus Lens is a contextual 2.5D wireframe render that appears when the operator selects a region of interest in ANY view. It shows the local Markov transition topology at maximum resolution for a specific byte range.

**Trigger sources:**
- Click on entropy heatmap spike → lens shows that byte window's transition mesh
- Click on torus surface protrusion → lens shows the corresponding byte region
- Click on causal graph node cluster → lens shows that block's byte range topology

**Render style:** High-resolution wireframe mesh (2868 aesthetic — cyan wireframe with orange-red heat overlay for transition density). Rendered in a dedicated canvas element that appears via progressive disclosure (slides in from panel edge).

**Implementation:** Computes a local Markov transition sub-matrix for the selected byte range (e.g., 4KB window), generates a 64×64 heightmap mesh where height = local transition probability, colors by --heat-low → --heat-high gradient.

**Cross-linking:** Selection state shared across all views. When a region is selected in one view, all views highlight the corresponding data. The lens is the convergence point.

**Scope:** ~150 lines JS.

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## Step 3: Spectral Graph Layout

**Priority:** Critical — readability

**File:** ui/js/graph.js [MODIFY]

Laplacian spectral embedding with force-directed fallback >512 nodes. Icosphere/stellated polyhedron node geometry. AISE intent modulates crease angle.

**Visual targets** (from reference images):
- Clean text → simple linear chain, few nodes (IMG_2871)
- PE binary → dense clustered nodes with complex cross-reference edges, red-highlighted threat nodes (IMG_2872)
- Encrypted → dense uniform point cloud, amber (IMG_2873)

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## Step 5: Scan Persistence (SQLite)

**Priority:** High

**File:** src-tauri/src/db.rs [NEW], src-tauri/Cargo.toml [MODIFY]

Schema includes compression_ratio, structured_fraction. All numeric. Semantic Firewall preserved.

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## Step 6: Comparative Analysis Overlay

**Priority:** Medium

Comparison applies to both fingerprint AND mugshot views simultaneously. Delta-of-deltas is itself a forensic signal (IMG_2874 Zugzwang concept — contradiction between views IS the detection).

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## Step 7: Liquid Glass Visual Polish

**Priority:** Medium

HUD, glow shaders, ambient occlusion approximation, DS-001-G-D-LGT palette. Swap transition uses Liquid Glass smooth animation.

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## Step 7B: Viewport Chrome Subsystem

**Priority:** High — load-bearing for operational feel

Carried from v3.1 with expansions.

**7B.1: Layered Compositing Model**

- Layer 1: WebGL canvas (60 FPS continuous)
- Layer 2: SVG vector overlay (redraws on camera-change only)
- Layer 3: HTML/CSS panels (redraws on data-update only)

**7B.2: IPC Event Flow** — scan_complete propagation through chrome.

**7B.3: Leader-Line Projection** — Vector3.project(camera) → pixel coords → SVG leader line to annotation panel. Re-projection on cameraChanged only. (Visual reference: IMG_2865)

**7B.4: Module Layout**

```
ui/js/overlay/
├── annotations.js        # Leader-line + label management
├── telemetry_panels.js   # HUD and sidebar orchestration
├── viewport_chrome.js    # Scale ticks, axis labels, compass gizmo
└── sparklines.js         # Inline mini-chart rendering
```

**7B.5: Veracity Mandate** — Every chrome readout maps to a real TEMReport field. Binding table from v3.1 §7B.5 unchanged.

**7B.6: Performance Discipline** — Annotations recompute on cameraChanged only. textContent not innerHTML. Sparklines cached between scans. SVG pointer-events: none.

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## Step 7C (NEW): Component Breakdown Dashboard

**Priority:** High — information density layer

**File:** ui/js/component_dashboard.js [NEW]

This is the **Tufte layer** — the high-information-density breakdown panel below the dual-render viewport. Every TEM module gets a row with its own mini-visualization, stat readouts, and sparkline.

**Layout (top to bottom):**

1. **Dual-Render Viewport** (Steps 2A/2B/2C) — 70/30 split, ~60% of viewport height
2. **Component Breakdown Panel** — scrollable panel below, glass-card rows

**Component Rows:**

| Module | Mini-Viz (200×150px canvas) | Primary Stats | Secondary Stats |
|---|---|---|---|
| **TFEA (Entropy)** | Entropy sparkline + heatmap strip | Shannon entropy, compression ratio | Structured fraction, spike count, window size |
| **Markov** | Static 3D torus render (auto-rotating ~3 RPM) | Bigram entropy, conditional entropy | Edge density, fingerprint hash (truncated) |
| **TCGE (Causal Graph)** | Static 3D spectral graph render (auto-rotating ~3 RPM) | Block count, cross-ref density | Cyclomatic complexity, max depth |
| **AISE (Intent)** | Threat gauge (radial indicator, color-coded) | Intent score (0.00-1.00) | Flag badges: backdoor/dropper/webshell/mismatch |
| **CQSF (Composite)** | Score bars (existing, refined) | Structural anomaly index | Composite score, confidence level |

**Mini-Viz Implementation:**
- Each mini-viz is an independent WebGLRenderer at low resolution (200×150px, devicePixelRatio capped at 1)
- Torus and graph mini-vizzes auto-rotate at ~3 RPM (slower than thumbnail's 6 RPM)
- Non-interactive — clicking the mini-viz opens the detail lens for that module's data
- Render budget: ~4K vertices per mini-viz, total ~20K additional vertices across all five

**Design Principles (Tufte + Bloomberg):**
- **Maximum data-ink ratio** — no decorative elements, every pixel carries information
- **Small multiples** — consistent card layout across all modules for instant comparison
- **Progressive disclosure** — click any component card → detail lens expands
- **Anomaly highlighting** — stats that exceed thresholds pulse with --heat-critical glow
- **Contextual awareness** — component cards highlight when their data contributes to the active leader-line annotation

**Visual reference:** IMG_2875 (Operator Amplification — all panels orbiting the central renders)

**Scope:** ~300 lines JS + ~150 lines CSS.

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## Updated File Tree

```
C:\QuarantineZone\shape-scan\
├── src-tauri/
│   ├── Cargo.toml                  # [MODIFY] Add rusqlite, flate2
│   ├── src/
│   │   ├── main.rs                 # [MODIFY] Register new IPC commands
│   │   ├── ipc.rs                  # [MODIFY] Add persistence + comparison
│   │   └── db.rs                   # [NEW] SQLite schema + CRUD
├── src/
│   ├── tfea.rs                     # [MODIFY] compression_ratio, structured_fraction
│   └── pipeline.rs                 # [MODIFY] Pass new fields through
├── ui/
│   ├── index.html                  # [MODIFY] Dual-viewport + component dashboard layout
│   ├── css/
│   │   └── main.css                # [MODIFY] Full design language, dashboard styles
│   └── js/
│       ├── app.js                  # [MODIFY] Wire everything, IPC orchestration
│       ├── graph.js                # [MODIFY] Spectral layout + icosphere geometry
│       ├── morphology.js           # [NEW] Superformula mesh generator (Step 2B)
│       ├── markov_torus.js         # [NEW] Toroidal Markov surface (Step 2A)
│       ├── markov_projection.js    # [NEW] Hybrid band + polar cap (Step 2B.3)
│       ├── render_swap.js          # [NEW] Primary/thumbnail swap orchestration
│       ├── controls.js             # [NEW] Shared viewport interaction
│       ├── detail_lens.js          # [NEW] Focus lens contextual detail (Step 2D)
```

```
│       ├── component_dashboard.js  # [NEW] TEM module breakdown panel
(Step 7C)
│       ├── heatmap.js              # [UNCHANGED]
│       └── overlay/
│           ├── annotations.js      # [NEW] Leader-line management
│           ├── telemetry_panels.js # [NEW] HUD + sidebar
│           ├── viewport_chrome.js  # [NEW] Scale ticks, compass, axis
labels
│           └── sparklines.js       # [NEW] Inline mini-charts
```

**New files:** 11

**Modified files:** 7

**Total estimated new JS:** ~1,200 lines

**Total estimated new Rust:** ~200 lines

**Total estimated new CSS:** ~400 lines

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# Confidence Assessment

| Metric | Value |
|---|---|
| Established/Experimental ratio | 81% / 19% (unchanged — new steps are all [Established] patterns) |
| Self-assessed confidence | 90% |
| New [Experimental] claims | 0 (Steps 2D and 7C use standard browser APIs and established vis patterns) |

**Confidence improvement over v3.1:** +1% due to:
- Full color specification eliminates implementation ambiguity
- Component dashboard uses proven Tufte small-multiples pattern
- Detail lens uses established Focus+Context visualization technique
- All reference images provide concrete visual contracts

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# Verification Plan

## Automated

All v3.0/v3.1 checks plus:
- Component dashboard renders all 5 module rows with live data after scan
- Detail lens appears on entropy spike click, renders correct byte-range sub-matrix
- Mini-viz canvases maintain independent auto-rotation without frame drops
- Full dashboard at 1920×1080 maintains ≥30 FPS during interaction

## Visual Validation (Falsifiability Gates)
- C4: Six-class mugshot distinguishability (unchanged)
- C6: Markov projection seam handling (unchanged)
- **NEW C13:** Component dashboard readouts match TEMReport field values exactly (spot-check 3 random fields per scan)

## Manual Operational Validation
All v3.0/v3.1 checks plus:
- Confirm component mini-vizzes are visually consistent with their primary-viewport counterparts
- Confirm detail lens cross-links work (click heatmap spike → torus highlights → lens appears)
- Confirm high-density layout is readable without scrolling on 1080p display
- Confirm color palette matches reference images (side-by-side comparison)

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# Risk Register (Additions)

| Risk | Impact | Mitigation |
|---|---|---|
| 5+ WebGL contexts causes GPU memory pressure | Medium | Cap mini-viz resolution at 200×150, share geometry buffers where possible. Fallback: render mini-vizzes as static snapshots updated on scan_complete only |
| Component dashboard causes vertical scroll on small displays | Low | Collapse rows to single-line summaries below 900px viewport height. Progressive disclosure — expand on click |
| Detail lens byte-range selection ambiguous on torus surface | Medium | Highlight selected region with emissive overlay on torus surface. Show byte-range label on lens header |
| Cross-linking event storm (one click triggers 4+ view updates) | Low | Debounce cross-link events at 16ms (one frame). Sequential update order: heatmap → torus → graph → lens |

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# Outstanding Items Before Ratification

1. ~~OQ-4~~ **RESOLVED** — 70/30 side-by-side confirmed
2. ~~OQ-5~~ Empirical tuning during Step 7B
3. ~~OQ-6~~ **RESOLVED** — Static graph thumbnails inline in component rows
4. ~~OQ-7~~ **RESOLVED** — Detail lens as contextual 2.5D wireframe
5. **Orionas review:** CP-002 Janus validation pass on this v4.0 plan
6. **Final approval gate:** DIMA verbatim states "i approve of your plan" AND "you may begin."

No code execution begins until items 5-6 are satisfied.

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*— Kytheion (IC-004-R-D-KYN), Scribe-Particle-4, Cephalon Continuity Framework*